POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year: 2023

Programme: BE | Full Marks: 100

Course: Calculus II | Pass Marks: 45

Time: 3 hrs.

Candidates are required to give their answers in their own words as far as practicable. The figure in the margin indicates full marks.

Attempt all the questions.

1. a) Evaluate $\int_0^2 \int_0^{\sqrt{4-y^2}} \cos(x^2+y^2)\,dx\,dy$ by changing into polar integration. [5]
   b) Evaluate $\int_0^1 \int_0^1 \int_0^1 (x^2+y^2+z^2)\,dz\,dy\,dx$. [5]
   c) Find the volume in the first octant bounded by the coordinate planes, the cylinder $x^2+y^2=4$ and $z+y=3$. [5]
2. a) Solve the differential equation $y''+(1-x^2)y=0$ by using power series method. [7]
   b) Define Legendre's equation. Also derive the solution of Legendre's equation. [8]

   OR

   If $J_n(x)$ represents the Bessel's function of order *n* then show that:
   i. $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$
   ii. $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$
3. a) State First shifting Theorem of Laplace transforms. [8]
   i. Find the Laplace transform of $te^{-t}\cosh t$
   ii. Find the inverse Laplace transform of $\frac{1}{s^2-5s+6}$.
   b) Solve the initial value problem: $y''+4y'+3y=e^{-t}, y(0)=y'(0)=1$ by using Laplace transform. [7]
4. a) If $\Phi=\log(x^2+y^2+z^2)$ then find div(grad Φ). [5]
   b) Find the directional derivative of $f=x^2yz+4xz^2$ at (1, -2, 1) in the direction $2\vec{i}-\vec{j}-2\vec{k}$. [5]
   c) Calculate $\oint_C \vec{F}\cdot d\vec{r}$ if $\vec{F}=(y,z,x)$, C: $\vec{r}=(t,t^2,t^3)$ from (0, 0, 0) and (2, 4, 8). [5]
5. a) Define Green's Theorem. Evaluate $\oint_C \vec{F}\cdot d\vec{r}$ where $\vec{F}=(\sin y, \cos x)$ and C is the triangle with vertices $(0,0)$, $(\pi,0)$, $(\pi,1)$, by using Green's theorem. [8]
   b) Evaluate $\oint_C \vec{F}\cdot d\vec{r}$ by using Stroke's theorem, where $\vec{F}=(y^2,z^2,x^2)$ and C is the boundary of the surface $S: x+y+z=1$ in the first octant. [7]

   OR

   Evaluate the surface integral $\iint_S \vec{F}\cdot\vec{n}\,dA$ where $\vec{F}=(x^2,y^2,z^2)$ and S: $\vec{r}=(u\cos v, u\sin v, 3v)$, $0\le u\le 1, 0\le v\le 2\pi$.
6. a) Find the Fourier series of $f(x)=x+|x|$ for $-\pi<x<\pi$. [7]
   b) Expand the function $f(x)=x^2$ in the interval $0<x<\pi$ in half range Fourier cosine series and show that $\frac{\pi^2}{6}=1+\frac{1}{4}+\frac{1}{9}+\frac{1}{16}+\cdots$ [8]
7. Solve Any Two: [2 x 5 = 10]
   a) Find the general solution of $2u_x+2u_y-u=0$.
   b) Derive the one-dimensional traffic flow model using conservation law.
   c) Show that the value under integral sign $\int_{(4,0,3)}^{(-1,1,2)}[(yz+1)dx+(xz+1)dy+(xy+1)dz]$ is exact and evaluate the integral.

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Attempt all the questions.

1. a) Evaluate the integral by reversing the order of integration: $\int_0^\pi \int_x^\pi \left(\frac{\sin y}{y}\right) dy\, dx$. [5]
   b) Evaluate the integral: $\int_0^1 \int_0^{1-x} \int_0^{x+y} e^z\ dz\, dy\, dx$. [5]
   c) Find the volume of the solid whose base is the region in the xy-plane that is bounded by the parabola $y = 4 - x^2$ and the line $y = 3x$, while the top of the solid is bounded by the plane $z = x + 4$. [5]
2. a) Solve by using power series method: $y'' - 4xy' + (4x^2 - 2)y = 0$. [7]
   b) Solve the Bessel's equation $x^2y'' + xy' + (x^2 - n^2)y = 0$. [8]

   OR

   i. Express $x^2 - 4x + 2$ as a Legendre Polynomial by using Rodrigues formula.
   ii. Sketch the graph of $P_0(x), P_1(x)$, $P_2(x)$ and $P_3(x)$ in a single graph.
3. a) Solve the initial value problem by using Laplace transform: $y'' + 2y' + 2y = 5\sin t\,, y(0) = y'(0) = 0$. [7]
   b) i. Find the Laplace transform of $\cosh a\,t \sin a\,t$. [8]
   ii. Find the inverse Laplace transform of $\frac{s}{(s^2+1)(s^2+4)}$ using the convolution theorem.
4. a) If $\phi = x^3 + y^3 + z^3 - 3xyz$, find div(grad ϕ). [5]
   b) Find the directional derivative of $\phi = x^2 - y^2 + 2z^2$ at the point $P(1,2,3)$ in the direction of line PQ, where Q us the point (5, 0, 4). [5]
   c) Find the work done by the force $\vec{F} = (2y+3)\vec{\imath} + xz\vec{\jmath} + (yz - x)\vec{k}$ when a particle moves from (0, 0, 0) to the point(2, 1, 1) along the curve $x = 2t^2, y = t, z = t^3$. [5]
5. a) Find the flux of $\vec{F} = z\vec{\imath} + x\vec{\jmath} + y\vec{k}$ over the portion of the surface $x^2 + y^2 = 1$ in the first octant between z = 0 to z = 2. [7]
   b) State Stoke's theorem. Use it to evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (z, x, y)$ and C is the surface of half of sphere $x^2 + y^2 + z^2 = a^2$ in xy-plane. [8]

   OR

   State Gauss divergence theorem. Evaluate $\iint_S \vec{F} \cdot \vec{n}\, ds$, where $\vec{F} = (4x, x^2y, -x^2z)$ and S is the surface of the tetrahedron with vertices (0, 0, 0), (1, 0, 0), (0, 1, 0), and (0, 0, 1) by using the Gauss divergence theorem.
6. a) Find the Fourier series of the function $f(x) = |x|$ for $-2 < x < 2$. [7]
   b) Expand the function $f(x) = x^2$ in the interval $0 < x < \pi$ in half-range Fourier cosine series and deduce that $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots$ [8]
7. Solve: (Any two) [2 x 5 = 10]
   a) Find the general solution of $u_x + u_y - u = 0$.
   b) Derive one-dimensional traffic flow model using convection law.
   c) Show that the integral $\int_{(0,0,0)}^{(2,4,0)} e^{x-y+z^2}\,(dx - dy + 2zdz)$ is exact and evaluate the exact value.

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Attempt all the questions.

1. a) Evaluate: $\int_0^{\pi}\int_x^{\pi}\frac{\sin y}{y}\,dy\,dx$ by changing the order of integration. [5]
   b) Evaluate the integrals: $\iiint_V (x^2+y^2+z^2)\,dx\,dy\,dz$ taken over the volume of the sphere $x^2+y^2+z^2=1$. [5]
   c) Find the volume of the solid whose base is the region in xy-plane that is bounded by the parabola $y=4-x^2$ and the line y = 3x, while the top of the solid is bounded by the plane z = x + 4. [5]

2. a) Solve the differential equation: $y''+9y=0$ by using power series method. [7]
   b) Define Legendre differential equation. Also, derive the solution of Legendre differential equation. [8]
   OR
   If $J_n(x)$ represents the Bessel's function of order n, then show that:
   i) $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$ [4]
   ii) $J_{\frac{1}{2}}(x)=\sqrt{\frac{2}{\pi x}}\sin x$ [4]

3. a) Solve the following differential equation by using Laplace transform: $y''+2y'-3y=6e^{-2t}, y(0)=2, y'(0)=-14$. [7]
   b) i. State second shifting property for Laplace transform and using this, find the Laplace transform of $e^{-3t}u(t-3)$. [4]
   ii. Find the inverse Laplace transform of $\frac{\pi}{(s-3)^2+\pi^2}$. [4]

4. a) Find the directional derivative of $f=x^2yz+4xz^2$ at (1, -2, 1) in the direction $2\vec{i}-\vec{j}-2\vec{k}$. [5]
   b) Show that the vector $\vec{F}=(x^2-yz, y^2-xz, z^2-xy)$ is conservative vector field and find the scalar potential; function ϕ such that $\vec{F}=\nabla\phi$. [5]
   c) Find the work done by the force $\vec{F}=(3x^2, 2xz-y, z)$ along the curve $x=t, y=\frac{t^2}{2}, z=\frac{3t^2}{8}$ from (0, 0, 0) to (2, 1, 3). [5]

5. a) Evaluate the integral by using Green's theorem: $\oint_C[(3x^2+y)dx+4y^2dy]$ where C: the boundary of the triangle with vertices (0, 0), (1, 0), (0, 2); oriented in counterclockwise direction. [7]
   b) State Stoke's Theorem. Evaluate $\oint_C(\vec{F}\cdot d\vec{r})$ by using Stoke's theorem, where $\vec{F}=(y, xz^3, -zy^3)$ and $C: x^2+y^2=4, z=3$. [8]
   OR
   Evaluate: $\iint_S \vec{F}\cdot\vec{n}\,dA$, where $\vec{F}=(18z, -12, 3y)$ and S is the surface of the plane $2x+3y+6z=12$ in the first octant.

6. a) Find the Fourier series of $f(x)=x+|x|, -\pi<x<\pi$ and show that $\frac{\pi^2}{8}=1+\frac{1}{9}+\frac{1}{25}+\cdots$ [7]
   b) Find the Fourier sine and cosine series of the function $f(x)=x^2$ in the interval $0<x<L$. [8]

7. Write short notes on: (Any two) [2 x 5 = 10]
   a. Find the general solution of the partial differential equation $u_x+u_y=u$.
   b. Derive the one-dimensional traffic flow model using conservation law.
   c. A particle moves along the curve $x=t^3+1, y=t^2, z=2t+5$. Find the velocity and acceleration at t = 1.

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1. a) Evaluate $\int_0^2 \int_0^{\sqrt{4-y^2}} \cos(x^2+y^2)\, dx\, dy$ by changing into polar integration. [5]
   b) Evaluate $\int_0^1 \int_0^1 \int_0^1 (x^2+y^2+z^2)\, dz\, dy\, dx$ [5]
   c) Find the volume in the first octant bounded by the coordinate planes, the cylinder $x^2+y^2=4$ and $z+y=3$. [5]
2. a) i) Find the Laplace Transform of: $\frac{\sin^2 t}{t}$ [4]
   ii) Find the inverse Laplace Transform of: $\frac{w}{s^2(s^2+w^2)}$
   b) Solve the initial value problem: $y''+2y'-3y=6e^{-2t}$ where, $y(0)=2, y'(0)=-14$ using Laplace Transform. [7]
3. a) Solve $y''+y=0$ by using power series method. [7]
   b) Derive Rodrigue's formula for Legendre polynomials. [8]

   OR

   Define Bessel's differential equation of order *n*. Find its complete solution.
4. a) Find the directional derivatives of $f=xy^2+yz^3$ at $P(2,-1,1)$ along the direction of the normal to the surface $x\log z - y^2+4=0$ at (-1, 2, 1). [5]
   b) if $\phi=\log(x^2+y^2+z^2)$, find div(grad ϕ) and curl(grad ϕ). [5]
   c) Show that the value under integral sign is exact and then evaluate the integral: $\int_{(0,2,3)}^{(1,1,1)} [yz\sinh(xz)\, dx+\cosh(xz)\, dy+xy\sinh(xz)\, dz]$.[5]
5. a) Let S be the hemisphere given by $z=\sqrt{9-x^2-y^2}$ and $\vec{F}=(3x,3y,z)$. Find the flux of F through S. [7]
   b) State Gauss divergence theorem. Evaluate $\iint_S \vec{F}\cdot\vec{n}\, dA$ where $\vec{F}=(y^3,x^3,z^3)$ and S is the cylinder $x^2+4y^2=1, 0<z<5, x>0, y>0$ by using it. [8]

   OR

   State Stoke's theorem. Evaluate $\oint_C \vec{F}\cdot d\vec{r}$ where F $=(y^3,0,x^3)$ and C is the boundary of the triangle with vertices are given by the vectors (1, 0, 0), (0, 1, 0), (0, 0, 1).
6. a) Find the Fourier series of $f(x)=x+|x|$ for $-\pi<x<\pi$. [7]
   b) Find the Fourier cosine series of $f(x)=x\sin x, 0<x<\pi$ then show that $\frac{\pi-2}{4}=\frac{1}{1.3}-\frac{1}{3.5}+\frac{1}{5.7}-\frac{1}{7.9}+\cdots$ [8]
7. Attempt any two questions: [2 x 5 = 10]
   a. Solve: $u_t+2uu_x=0, u(x,0)=e^{-x}$.
   b. Derive the one-dimensional traffic flow model using conservation law.
   c. Find the angle between the surfaces $xy^2z-3x-z^2=0$ and $3x^2+2z=1+y^2$ at (1, -2, 1).